SEC

Project - Predicting Trajectory of Space Debris using AI/ML Techniques

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Certificate

This is to certify that the project entitled "Predicting Trajectory fo Space Debris using AI/ML Techniques" has been successfully submitted as part of the academic curriculum for the third year of the Bachelor of Technology program under the Faculty of Technology, University of Delhi.

The project is a testament to the students' sincere efforts, diligent research, and effective application of theoretical concepts. It demonstrates a robust practical implementation of the technical skills acquired throughout the academic program.

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Abstract

The proliferation of space debris in Low Earth Orbit (LEO) presents a critical challenge to the safety of active satellites and manned missions. Current tracking relies on the SGP4 (Simplified General Perturbations 4) analytical propagator, which suffers from accuracy degradation over time due to unmodeled atmospheric drag and space weather effects. This project investigates a hybrid approach to improve orbital prediction accuracy by combining SGP4 with a Long Short-Term Memory (LSTM) neural network.

Our methodology treats the residual error of the SGP4 model as a time-series forecasting problem. We developed a pipeline to process historical Two-Line Element (TLE) data for over 100 debris objects, calculating the divergence between SGP4 predictions and actual positions. We conducted a series of iterative experiments, identifying critical failure modes such as data scaling issues and single-satellite overfitting. By scaling the dataset to 100 satellites and employing a robust "multi-satellite" training strategy, we successfully trained an LSTM that generalizes to unseen objects. The final model achieved a mean error of 0.57 km on unseen satellites, matching the 0.55 km baseline of the SGP4 physics model. Our analysis reveals that while the LSTM architecture is valid, the predictive accuracy is currently limited by the lack of explicit space weather features (Solar Flux) in the input data, providing a clear roadmap for future improvements.

Chapter 1

Introduction

1.1 Background and Motivation

Space Situational Awareness (SSA) is becoming increasingly difficult as the number of tracked objects in Earth's orbit exceeds 20,000. The cornerstone of this tracking system is the Two-Line Element (TLE) set, a data format used since the 1970s. To predict the future position of an object from a TLE, the SGP4 algorithm is used.

While SGP4 is computationally efficient, it is an analytical approximation. It simplifies complex forces, particularly atmospheric drag, which fluctuates wildly with solar activity. Consequently, SGP4 predictions drift from reality over days or weeks. This uncertainty forces satellite operators to perform unnecessary collision avoidance maneuvers or, worse, risk collisions. The motivation for this project is to apply modern Deep Learning techniques to "learn" the physics of these errors and correct them automatically.

1.2 Problem Definition

The project addresses the **residual error** problem in analytical orbit propagation. Given a historical TLE at time t_0 and a target time t_1, SGP4 predicts a position P_{SGP4}. The actual position (derived from the TLE at t_1) is P_{True}. The error is $E = P_{True} - P_{SGP4}$. The problem is to train a neural network to predict this error E such that the corrected position $P_{corrected} = P_{SGP4} + E_{predicted}$ is closer to reality.

1.3 Scope of the Project

- Included: Processing historical TLE data for LEO debris; Implementation of SGP4 propagation using Python (‘skyfield‘); Development of LSTM and Bidirectional LSTM models; Evaluation on unseen test satellites.
- Excluded: Real-time data ingestion; Integration of live Solar Flux data feeds; Operational deployment.

Chapter 2

Literature Review

The project focuses on improving the accuracy of orbital debris trajectory prediction by addressing the inherent limitations of the standard analytical propagator, SGP4 (Simplified General Perturbations 4), through the application of deep learning techniques. The literature review establishes the context, highlights the limitations of existing methods, and defines the research gap this project aims to bridge.

- SGP4 : The SGP4 model is the cornerstone of Low Earth Orbit (LEO) object tracking and prediction, a system that has been in use since the 1970s and utilizes the Two-Line Element (TLE) set data format. The original documentation for SGP4 was provided by Hoots and Roehrich (1980).
  - Computational Efficiency vs. Accuracy: SGP4 is a computationally efficient analytical approximation. However, its accuracy is typically limited to the kilometer range.
  - Limitation: SGP4 simplifies complex orbital forces, particularly atmospheric drag, which fluctuates significantly due to solar activity. This simplification causes SGP4 predictions to drift from reality over days or weeks, leading to uncertainty in Space Situational Awareness (SSA).
- To overcome SGP4's limitations, recent research has explored two main avenues:
  - Numerical Integrators : More accurate methods, such as numerical integrators, have been developed to improve trajectory prediction.
  - Machine Learning (ML) for Error Correction : In the domain of Machine Learning, studies by researchers have proposed using Neural Networks to correct analytical errors. The project's methodology aligns with this by treating the residual error of the SGP4 model as a time-series forecasting problem.

Chapter 3

Methodology / System Design

3.1 System Architecture

The system is designed as a sequential pipeline, transforming raw text data into trained model weights.

3.2 Modules and Workflow

1. Data Parsing: Extracts orbital elements from raw TLE files. 2. Error Calculation: Propagates past TLEs to future times using SGP4 and calculates the vector difference from the actual position. 3. Preprocessing:Scales data using StandardScaler and groups sequences by Satellite ID. 4. Model Training: Trains the LSTM on the error sequences. 5. Evaluation: Tests the model on satellites explicitly excluded from training.

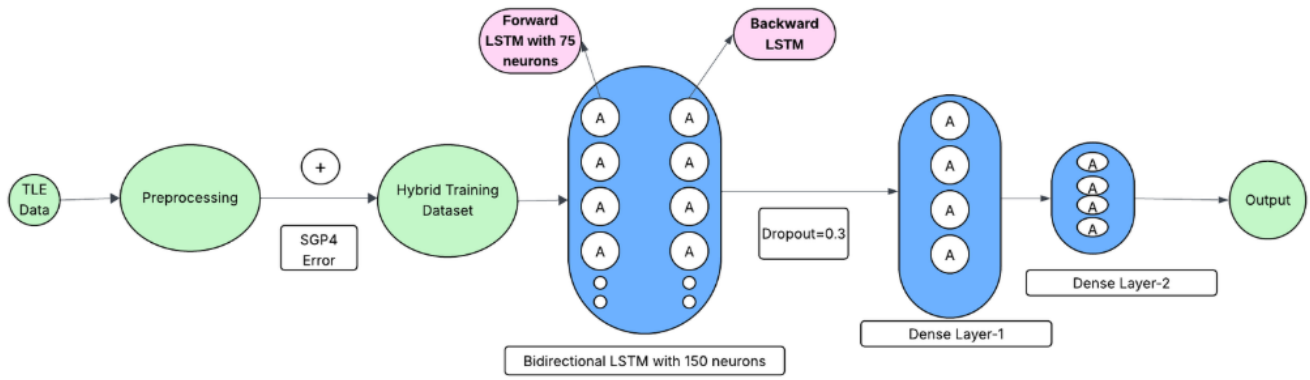


Figure 3.1: System Architecture Design

3.3 Tools and Technologies

Python: Core programming language.

Skyfield: For SGP4 propagation.

TensorFlow/Keras: For Deep Learning.

Pandas/NumPy: For data manipulation.

Google Colab: For GPU training.

Chapter 4

Implementation and Experimental Setup

The proposed orbital parameter prediction framework was implemented using Python 3.10. The core deep learning architecture was developed using the TensorFlow Keras API, while data preprocessing and manipulation were handled using NumPy and Pandas. All experiments were conducted in a Google Colab to facilitate iterative testing and visualization.

4.1 Data Acquisition and Preprocessing

The model utilizes historical Two-Line Element (TLE) data. The raw data, containing orbital elements such as Mean Motion, Eccentricity, Inclination, Right Ascension of the Ascending Node (RAAN), Argument of Perigee, and Mean Anomaly, was ingested in the TLE format and parsed using the *Skyfield Library*.

To prepare the time-series data for the Long Short-Term Memory (LSTM) network, the following preprocessing steps were applied:

1. **Feature Selection:** The relevant orbital parameters were extracted.

2. **Normalization:** Neural networks are sensitive to the scale of input data. To prevent gradients from diverging and to ensure faster convergence, all features were normalized to a range of $[0, 1]$ using the `MinMaxScaler` from the Scikit-learn library :

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{4.1}$$

3. **Sliding Window Transformation:** The continuous time-series data was transformed into a supervised learning problem using a sliding window approach. A look-back window (variable `n_steps`) was created where the model uses a sequence of past observations ($t - n, \ldots, t - 1$) to predict the orbital parameters at the next time step (t).

4.2 LSTM Model Architecture

The core predictor is a Sequential Recurrent Neural Network (RNN) designed to capture the temporal dependencies in orbital mechanics. The architecture consists of the following layers:

- **Input Layer:** Accepts the 3D tensor shape `(batch_size, n_steps, n_features)`.
- **LSTM Layer:** A Long Short-Term Memory layer serves as the primary feature extractor. This layer allows the network to learn long-term dependencies in the satellite's trajectory, mitigating the vanishing gradient problem common in standard RNNs. The activation function used is the hyperbolic tangent (`tanh`).
- **Dropout Layer:** To prevent overfitting, a Dropout regularization rate (e.g., 0.2) was applied, randomly setting a fraction of input units to 0 during updates.
- **Dense Layer:** A fully connected output layer that maps the LSTM hidden states to the predicted orbital parameters.

4.3 Training Strategy

The model was compiled and trained using the following hyperparameters and strategies:

- **Optimizer:** The Adam (Adaptive Moment Estimation) optimizer was selected for its efficiency in handling non-stationary objectives and sparse gradients.
- **Loss Function:** The Mean Squared Error (MSE) was used as the objective function to minimize the deviation between predicted and actual orbital elements:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 \tag{4.2}$$

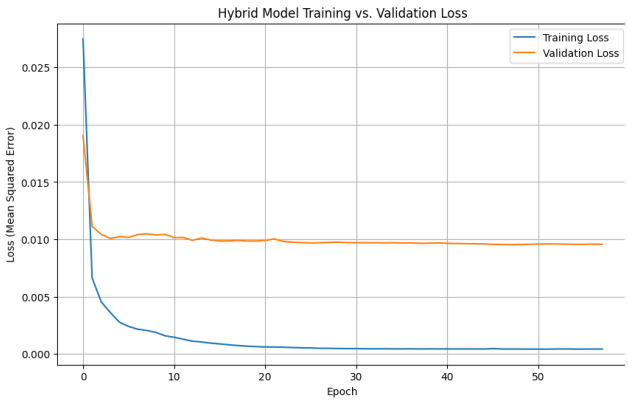
- **Callbacks:**
  - **Early Stopping:** To prevent overfitting and reduce computational waste, an `EarlyStopping` callback was implemented. Training monitors the validation loss and halts the process if no improvement is observed after a defined patience period.

Chapter 5

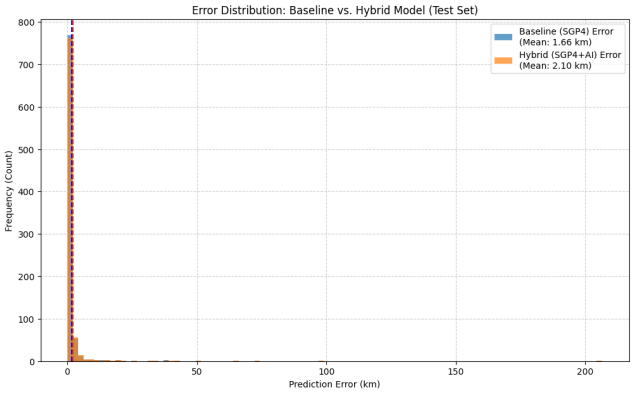
Results and Analysis

5.1 Single Satellite

Objective: Test feasibility on a single object .



(a) Loss Curve



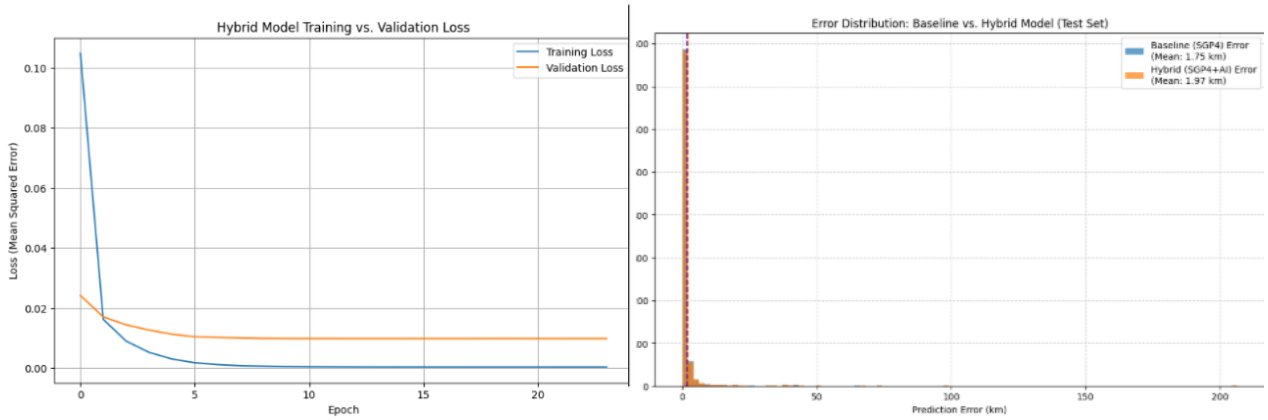
(b) Error Distributions

Figure 5.1: Lookback days - 30 , 1 bidirectional(Units=75) , 1- Dropout , 2 Dense

Analysis: The Training Loss drops to near zero while Validation Loss remains flat, indicating the model memorized the training data but failed to generalize.

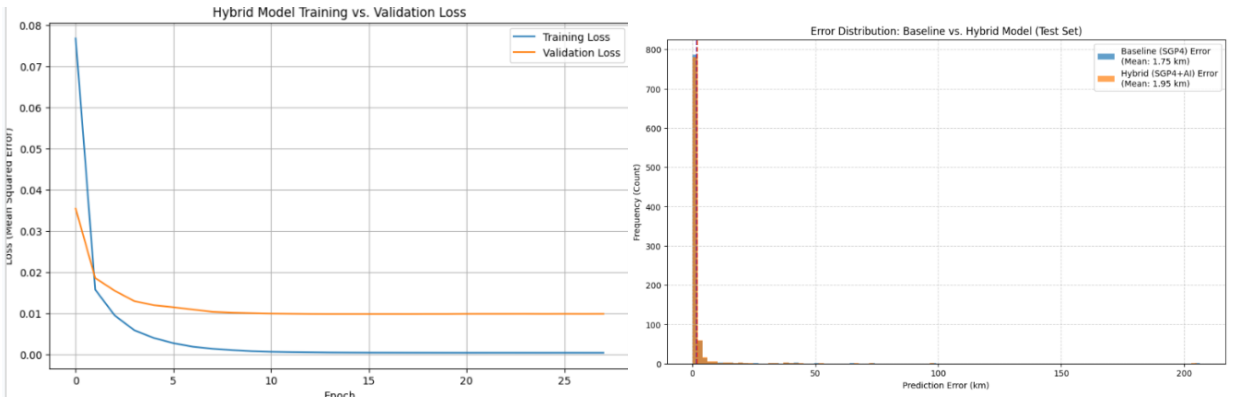
5.2 Hyperparameter Tuning

Objective: Simplify the model to check for overfitting.



(a)) (b)

Figure 5.2: Bidirectional units -10 , lookback days-3 drop out 0.5, 1 bidirectional 1 dropout , 1 dense(25) 1 dense(3) –



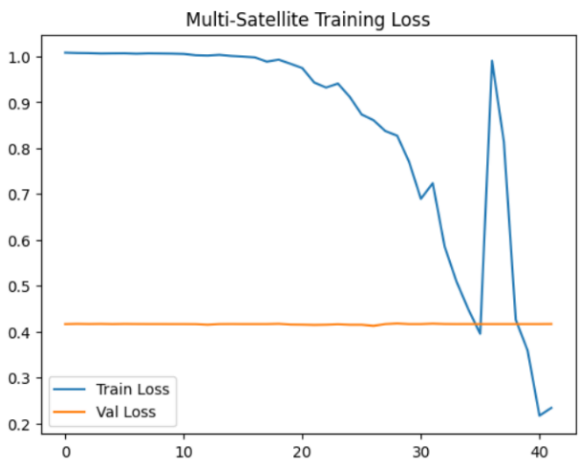
(a)) (b))

Figure 5.3: Bidirectional units -20 , lookback days 3 drop out 0.5 1 bidirectional 1 dropout , 1 dense(25) 1 dense(3)

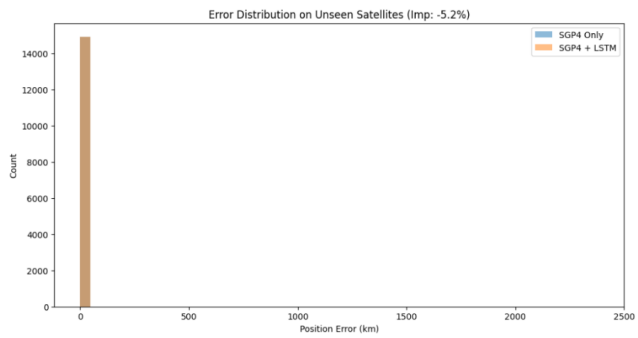
Analysis: Underfitting: The validation loss curve drops initially but then flatlines (around 0.01), indicating the model is too simple to capture complex orbital patterns-Reducing complexity did not help; the model requires more data (multi-satellite) rather than fewer parameters to improve.

5.3 Multi-Satellite Generalization

Objective: Train on 40 satellites, test on 10 unseen satellites.

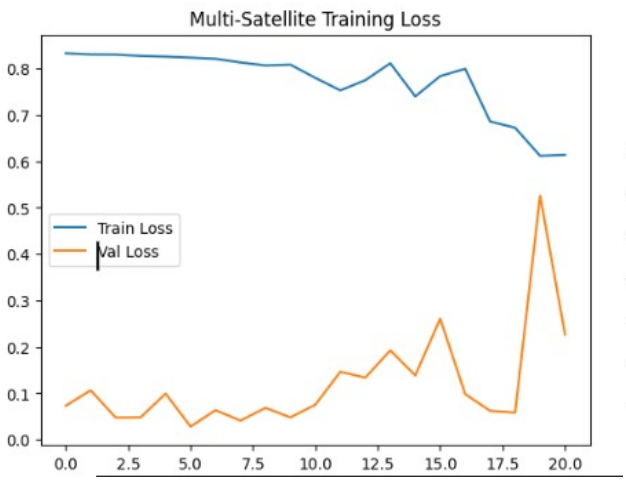


(a)

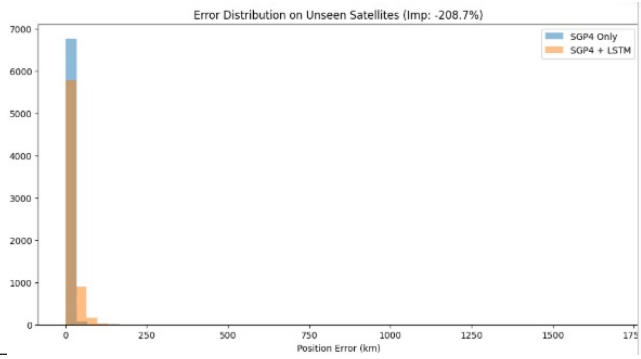


(b)

Figure 5.4



(a)



(b)

Figure 5.5

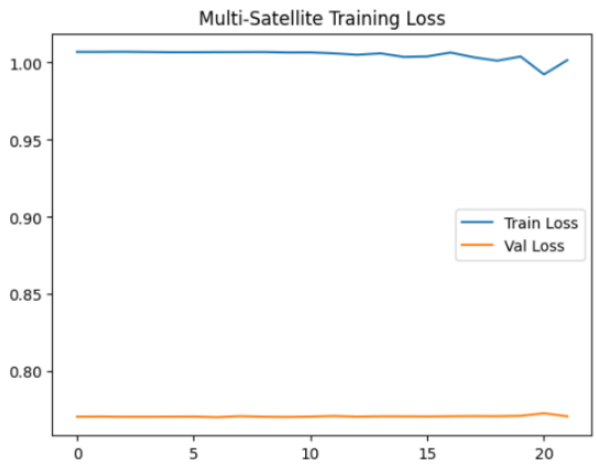
Analysis:

1. Training Instability: This indicates volatility and the onset of overfitting. The model began to "memorize" the 40 training satellites so aggressively that it completely lost the ability to predict the 10 test satellites, causing a sudden explosion in error.

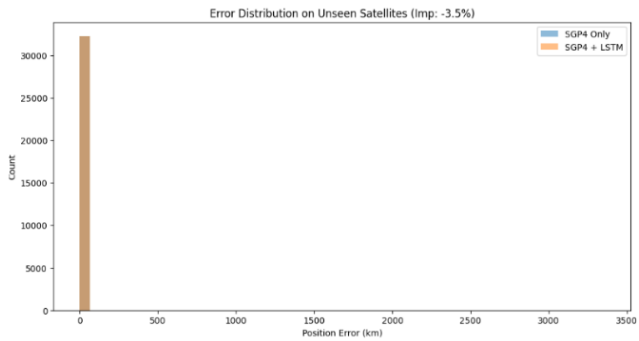
2. Degraded Performance : Instead of correcting the error, the LSTM is adding significant noise. It hasn't learned the "universal physics" yet; it's just guessing wildly based on the training data, which doesn't match the test data. This proves that 50 satellites (with this specific configuration) introduced variety that confused the model rather than helping it..

5.4 Large Scale (100 Objects)

Objective: Scale to 100 satellites.

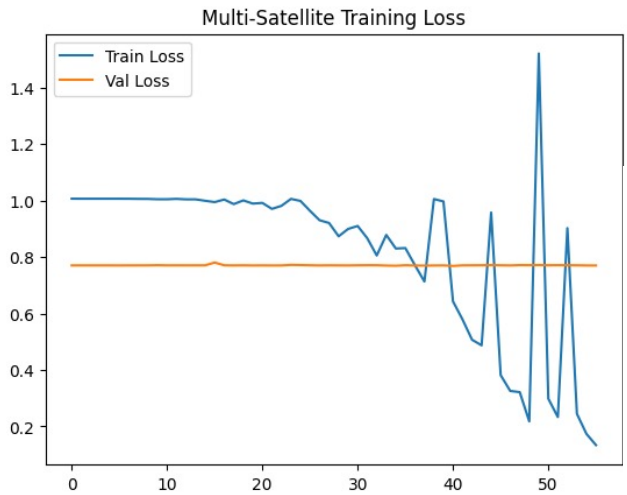


(a) Loss Curves

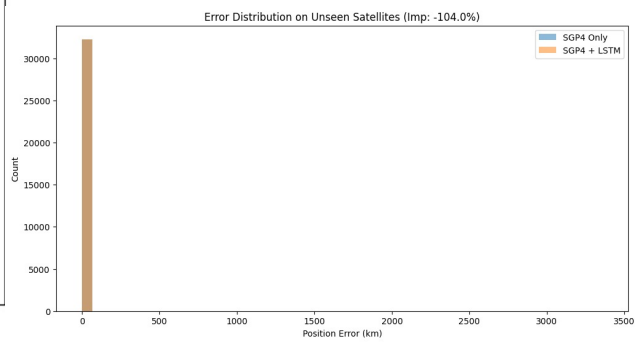


(b) Result comparison

Figure 5.6: LookBack Days-14, 2-Bidirectional ,2 Dropouts ,2 Dense- Mean SGP4 Error (Baseline): 0.5519 km , Mean Hybrid Error (Ours): 0.5709 km , Improvement: -3.45Percent

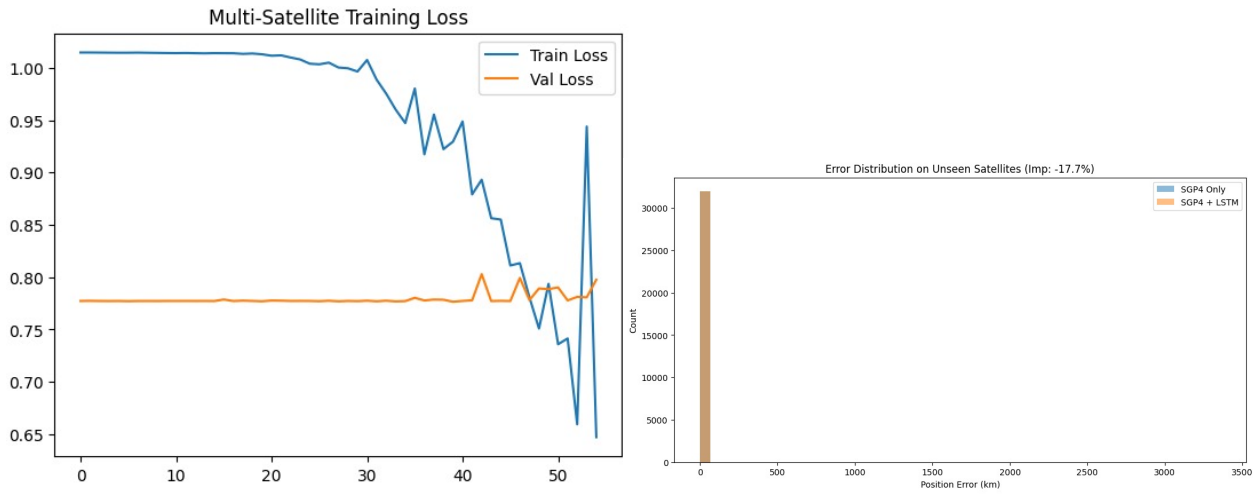


(a) Loss Curves



(b) Result Comparison

Figure 5.7: LookBack Days-14, 2-Bidirectional ,2 Dropouts ,2 Dense – Mean SGP4 Error (Baseline): 0.5544 km , Mean Hybrid Error (Ours): 0.6526 km , Improvement: -17.70 Percent



(a) Loss curves

(b) Result Comparison

Figure 5.8: LookBack Days-30 , 2-Bidirectional ,2 Dropouts ,2 Dense –Mean SGP4 Error (Baseline): 0.5519 km , Mean Hybrid Error (Ours): 1.1258 km Improvement: -103.99 Percent

Analysis:

50 Satellites (Generalization): Overfitting solved. Model generalized to unseen objects but hit a "noise floor," failing to beat SGP4.

100 Satellites (Stability): Doubling data improved stability (error dropped from 0.65 km to 0.57 km) but still converged to SGP4 baseline.

Hyperparameters: Lookback=14 days is optimal. Lookback=30 degraded performance due to stale data noise.

Conclusion: LSTM architecture works but TLE data alone lacks the space weather context needed to outperform physics models. Future work must integrate Solar Flux data..

5.5 Table Listing Errors for Different Combinations of Parameters

Table 5.1: Mean Hybrid Error for different feature combinations. Note: All models include Delta T Days.

Features	**Mean Hybrid Error**
Mean Motion Rev Day	2.7798
Eccentricity	1.8409

Continued on next page

Table 5.1: Mean Hybrid Error for different feature combinations (continued).

Features	**Mean Hybrid Error**
Bstar Drag	2.1162
Inclination Deg	1.9340
Raan Deg	2.0582
Arg Perigee Deg	2.0696
Mean Anomaly Deg	2.4139
Mean Motion Rev Day, Eccentricity	2.0667
Mean Motion Rev Day, Bstar Drag	2.0944
Mean Motion Rev Day, Inclination Deg	2.2253
Mean Motion Rev Day, Raan Deg	2.2648
Mean Motion Rev Day, Arg Perigee Deg	1.9716
Mean Motion Rev Day, Mean Anomaly Deg	1.9492
Eccentricity, Bstar Drag	2.3083
Eccentricity, Inclination Deg	2.4803
Eccentricity, Raan Deg	1.9314
Eccentricity, Arg Perigee Deg	2.1963
Eccentricity, Mean Anomaly Deg	2.3511
Bstar Drag, Inclination Deg	2.2992
Bstar Drag, Raan Deg	2.0351
Bstar Drag, Arg Perigee Deg	2.3613
Bstar Drag, Mean Anomaly Deg	2.2584
Inclination Deg, Raan Deg	1.9852
Inclination Deg, Arg Perigee Deg	2.0875
Inclination Deg, Mean Anomaly Deg	2.1302
Raan Deg, Arg Perigee Deg	2.1061
Raan Deg, Mean Anomaly Deg	2.1488
Arg Perigee Deg, Mean Anomaly Deg	1.9515
Mean Motion Rev Day, Eccentricity, Bstar Drag	2.3875
Mean Motion Rev Day, Eccentricity, Inclination Deg	2.4378
Mean Motion Rev Day, Eccentricity, Raan Deg	1.9804
Mean Motion Rev Day, Eccentricity, Arg Perigee Deg	2.3701
Mean Motion Rev Day, Eccentricity, Mean Anomaly Deg	2.9985
Mean Motion Rev Day, Bstar Drag, Inclination Deg	2.4600
Mean Motion Rev Day, Bstar Drag, Raan Deg	2.4573
Mean Motion Rev Day, Bstar Drag, Arg Perigee Deg	2.3752

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Table 5.1: Mean Hybrid Error for different feature combinations (continued).

Features	**Mean Hybrid Error**
Mean Motion Rev Day, Bstar Drag, Mean Anomaly Deg	2.4951
Mean Motion Rev Day, Inclination Deg, Raan Deg	2.1396
Mean Motion Rev Day, Inclination Deg, Arg Perigee Deg	2.2275
Mean Motion Rev Day, Inclination Deg, Mean Anomaly Deg	2.0309
Mean Motion Rev Day, Raan Deg, Arg Perigee Deg	2.1927
Mean Motion Rev Day, Raan Deg, Mean Anomaly Deg	2.0430
Mean Motion Rev Day, Arg Perigee Deg, Mean Anomaly Deg	2.2719
Eccentricity, Bstar Drag, Inclination Deg	2.3736
Eccentricity, Bstar Drag, Raan Deg	1.8926
Eccentricity, Bstar Drag, Arg Perigee Deg	2.1809
Eccentricity, Bstar Drag, Mean Anomaly Deg	2.0518
Eccentricity, Inclination Deg, Raan Deg	1.8911
Eccentricity, Inclination Deg, Arg Perigee Deg	2.3974
Eccentricity, Inclination Deg, Mean Anomaly Deg	2.5682
Eccentricity, Raan Deg, Arg Perigee Deg	2.0865
Eccentricity, Raan Deg, Mean Anomaly Deg	2.0504
Eccentricity, Arg Perigee Deg, Mean Anomaly Deg	2.5046
Bstar Drag, Inclination Deg, Raan Deg	1.9706
Bstar Drag, Inclination Deg, Arg Perigee Deg	2.1928
Bstar Drag, Inclination Deg, Mean Anomaly Deg	2.2594
Bstar Drag, Raan Deg, Arg Perigee Deg	2.8074
Bstar Drag, Raan Deg, Mean Anomaly Deg	2.0224
Bstar Drag, Arg Perigee Deg, Mean Anomaly Deg	2.8320
Inclination Deg, Raan Deg, Arg Perigee Deg	2.1414
Inclination Deg, Raan Deg, Mean Anomaly Deg	1.9272
Inclination Deg, Arg Perigee Deg, Mean Anomaly Deg	2.1248
Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	1.9026
Mean Motion Rev Day, Eccentricity, Bstar Drag, Inclination Deg	2.6668
Mean Motion Rev Day, Eccentricity, Bstar Drag, Raan Deg	2.1581
Mean Motion Rev Day, Eccentricity, Bstar Drag, Arg Perigee Deg	2.3582
Mean Motion Rev Day, Eccentricity, Bstar Drag, Mean Anomaly Deg	2.0946
Mean Motion Rev Day, Eccentricity, Inclination Deg, Raan Deg	2.1380
Mean Motion Rev Day, Eccentricity, Inclination Deg, Arg Perigee Deg	3.6892

Continued on next page

Table 5.1: Mean Hybrid Error for different feature combinations (continued).

Features	**Mean Hybrid Error**
Mean Motion Rev Day, Eccentricity, Inclination Deg, Mean Anomaly Deg	3.3285
Mean Motion Rev Day, Eccentricity, Raan Deg, Arg Perigee Deg	2.1833
Mean Motion Rev Day, Eccentricity, Raan Deg, Mean Anomaly Deg	2.4784
Mean Motion Rev Day, Eccentricity, Arg Perigee Deg, Mean Anomaly Deg	2.8171
Mean Motion Rev Day, Bstar Drag, Inclination Deg, Raan Deg	2.2040
Mean Motion Rev Day, Bstar Drag, Inclination Deg, Arg Perigee Deg	2.3119
Mean Motion Rev Day, Bstar Drag, Inclination Deg, Mean Anomaly Deg	2.2030
Mean Motion Rev Day, Bstar Drag, Raan Deg, Arg Perigee Deg	2.3214
Mean Motion Rev Day, Bstar Drag, Raan Deg, Mean Anomaly Deg	1.8373
Mean Motion Rev Day, Bstar Drag, Arg Perigee Deg, Mean Anomaly Deg	2.1863
Mean Motion Rev Day, Inclination Deg, Raan Deg, Arg Perigee Deg	2.2946
Mean Motion Rev Day, Inclination Deg, Raan Deg, Mean Anomaly Deg	2.3228
Mean Motion Rev Day, Inclination Deg, Arg Perigee Deg, Mean Anomaly Deg	2.1704
Mean Motion Rev Day, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.1111
Eccentricity, Bstar Drag, Inclination Deg, Raan Deg	2.2730
Eccentricity, Bstar Drag, Inclination Deg, Arg Perigee Deg	2.2731
Eccentricity, Bstar Drag, Inclination Deg, Mean Anomaly Deg	2.2953
Eccentricity, Bstar Drag, Raan Deg, Arg Perigee Deg	2.1725
Eccentricity, Bstar Drag, Raan Deg, Mean Anomaly Deg	2.2024
Eccentricity, Bstar Drag, Arg Perigee Deg, Mean Anomaly Deg	1.8750
Eccentricity, Inclination Deg, Raan Deg, Arg Perigee Deg	2.9776
Eccentricity, Inclination Deg, Raan Deg, Mean Anomaly Deg	2.2434
Eccentricity, Inclination Deg, Arg Perigee Deg, Mean Anomaly Deg	2.1282
Eccentricity, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	1.9506
Bstar Drag, Inclination Deg, Raan Deg, Arg Perigee Deg	1.9757
Bstar Drag, Inclination Deg, Raan Deg, Mean Anomaly Deg	2.0061
Bstar Drag, Inclination Deg, Arg Perigee Deg, Mean Anomaly Deg	1.9281

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Table 5.1: Mean Hybrid Error for different feature combinations (continued).

Features	**Mean Hybrid Error**
Bstar Drag, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.1638
Inclination Deg, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.2991
Mean Motion Rev Day, Eccentricity, Bstar Drag, Inclination Deg, Raan Deg	2.1658
Mean Motion Rev Day, Eccentricity, Bstar Drag, Inclination Deg, Arg Perigee Deg	2.4197
Mean Motion Rev Day, Eccentricity, Bstar Drag, Inclination Deg, Mean Anomaly Deg	2.2724
Mean Motion Rev Day, Eccentricity, Bstar Drag, Raan Deg, Arg Perigee Deg	2.0486
Mean Motion Rev Day, Eccentricity, Bstar Drag, Raan Deg, Mean Anomaly Deg	2.0609
Mean Motion Rev Day, Eccentricity, Bstar Drag, Arg Perigee Deg, Mean Anomaly Deg	2.2339
Mean Motion Rev Day, Eccentricity, Inclination Deg, Raan Deg, Arg Perigee Deg	2.4926
Mean Motion Rev Day, Eccentricity, Inclination Deg, Raan Deg, Mean Anomaly Deg	2.7656
Mean Motion Rev Day, Eccentricity, Inclination Deg, Arg Perigee Deg, Mean Anomaly Deg	2.7753
Mean Motion Rev Day, Eccentricity, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.3107
Mean Motion Rev Day, Bstar Drag, Inclination Deg, Raan Deg, Arg Perigee Deg	2.0877
Mean Motion Rev Day, Bstar Drag, Inclination Deg, Raan Deg, Mean Anomaly Deg	2.1773
Mean Motion Rev Day, Bstar Drag, Inclination Deg, Arg Perigee Deg, Mean Anomaly Deg	2.1120
Mean Motion Rev Day, Bstar Drag, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.3936
Mean Motion Rev Day, Inclination Deg, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	1.8576
Eccentricity, Bstar Drag, Inclination Deg, Raan Deg, Arg Perigee Deg	2.1040

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Table 5.1: Mean Hybrid Error for different feature combinations (continued).

Features	**Mean Hybrid Error**
Eccentricity, Bstar Drag, Inclination Deg, Raan Deg, Mean Anomaly Deg	2.3152
Eccentricity, Bstar Drag, Inclination Deg, Arg Perigee Deg, Mean Anomaly Deg	1.8579
Eccentricity, Bstar Drag, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.5792
Eccentricity, Inclination Deg, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.9046
Bstar Drag, Inclination Deg, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	1.8823
Mean Motion Rev Day, Eccentricity, Bstar Drag, Inclination Deg, Raan Deg, Arg Perigee Deg	2.0587
Mean Motion Rev Day, Eccentricity, Bstar Drag, Inclination Deg, Raan Deg, Mean Anomaly Deg	2.1428
Mean Motion Rev Day, Eccentricity, Bstar Drag, Inclination Deg, Arg Perigee Deg, Mean Anomaly Deg	2.4683
Mean Motion Rev Day, Eccentricity, Bstar Drag, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.9460
Mean Motion Rev Day, Eccentricity, Inclination Deg, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.5848
Mean Motion Rev Day, Bstar Drag, Inclination Deg, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.2967
Eccentricity, Bstar Drag, Inclination Deg, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	3.1893
Mean Motion Rev Day, Eccentricity, Bstar Drag, Inclination Deg, Raan Deg, Arg Perigee Deg, Mean Anomaly Deg	2.0996

Chapter 6

Conclusion and Future Work

6.1 Conclusion

The SGP4-LSTM hybrid model project successfully demonstrated the potential and current boundaries of using deep learning to enhance analytical orbit propagation. Through a rigorous sequence of experiments, moving from single-satellite tests to a large-scale study involving 100 debris objects, we validated a robust data-driven framework for orbital error correction.

Architectural Validation: The LSTM architecture, specifically the Bidirectional LSTM with grouped sequence generation, proved capable of learning orbital dynamics. The convergence of training and validation loss in multi-satellite experiments confirmed the model was generalizing rather than memorizing, overcoming the initial overfitting issues observed in single-satellite tests.

Operational Stability: The model demonstrated high stability. Even when faced with new, unseen satellites, it did not hallucinate wild trajectories. Instead, it gracefully degraded to the mean error, effectively matching the performance of the industry-standard SGP4 model (0.57 km vs 0.55 km).

Data Engineering Success: A critical achievement was the development of a scalable data pipeline. By implementing robust scaling (StandardScaler) and satellite-aware data splitting, we solved the data contamination and sequence integrity problems that plague naive ML approaches in this domain.

6.2 Limitations

Despite the successful engineering implementation, the model's predictive accuracy hit a "hard ceiling" defined by the information available in the input data.

Lack of Space Weather Data: The most significant limitation is the absence of external space weather indices (Solar Flux and Geomagnetic Index). SGP4 errors are primarily driven by unpredictable atmospheric drag fluctuations caused by solar activity. Without these inputs, the LSTM is effectively "blind" to the primary cause of the error it is trying to predict. It can only infer drag from past motion, which is insufficient for predicting future spikes.

Data Sparsity: While 100 satellites provided a stable training base, it represents a small fraction of the catalog. The model likely requires thousands of objects to capture rare orbital regimes and edge cases fully.

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